Radar | Complete Illustrated Research Report

GEMSTAR Task 23: reflection fields, TOF decoding, startup behavior, 3D geometry, MATLAB checks, controlled 2 m displacement and OSI serialization.

Evidence source: uploaded Task_23_Final_Research.zip. Figures and tables come from archived assets. This document explains established findings separately from assumptions and future validation.

Executive research conclusions

Radar work covers OSI reflection availability, TOF/range/Doppler-field inspection, startup anomaly diagnostics, 3D coordinate reconstruction, MATLAB numerical cross-checks, tentative LiDAR proximity comparisons and a controlled cube displacement. Task 23.13 directly checked that OSI-recorded TOF preserves the prior static geometric range change.

Measure	Baseline	Displaced
Static geometric range	8.534070 m	10.542090 m
OSI-derived range	8.536549 m	10.545149 m
Absolute difference	2.479 mm	3.059 mm
Decoded minus geometric change	0.580 mm	0.580 mm (difference metric)

How each Radar subtask contributed

23.5A/C.1-2 | Radar reflection inventory and startup

Inspect Radar configuration and reflection fields including time_of_flight and signal_strength, count usable measurements, and flag startup frames with anomalous returns rather than assuming all frames are stable.

23.5C.3/C.4 | Radar 3D geometry and MATLAB

Recover ranges from TOF, construct local x=r*cos(el)*cos(az), y_UE=r*cos(el)*sin(az), z=r*sin(el), flip Unreal Y for OSI and add recorded mount translation for zero rotation. MATLAB cross-check tests numerical parity, not external accuracy.

23.5D.1-2 | Radar-LiDAR overlay and timestamp

Overlay sampled points in a common vehicle-OSI coordinate convention. The nearest-neighbour P95 distances can be large for selected frames; shared SensorView timestamps mean shared container time, not independent acquisition alignment.

23.5D.3-6 | Exploratory candidate association

Vary distance thresholds, actor-ID consistency rules, candidate rejections and mutual constraints. Graphs illustrate sensitivity of hypothetical correspondences. They are preliminary experiments, NOT a validated Task 24 fusion result.

23.6-7 | Controlled calibration design and persistence

Research map/actor checks and multiple iterations of cube placement. The final 2 m target movement only became reliable after map edits were persisted and independently reloaded. Do not count failed setup attempts as successful calibration.

23.7AS | Independent geometric displacement

Four central rays (44,45,54,55) give baseline 8.534070 m, displaced 10.542090 m, so a 2 m target shift produces 2.008020 m ray-path change owing to angled rays.

23.13B | OSI serialization test

Decode 4-byte little-endian length-prefixed SensorView messages; examine first 10 frames per condition and four rays, 80 positive samples. TOF-derived ranges are 8.536549 m and 10.545149 m, with 0.579953 mm change discrepancy relative to geometric range changes.

23.13C/D | Robustness and publication-ready evidence

Test two valid and ten invalid reflection input cases (12/12); make Radar serialization plot, evidence manifest and final scope-limited report.

23.15 | Unified Radar adapter

Extract original validated preprocess_radar_return without executing experiment module, preserve status/range/strength and timestamps; 13/13 integration tests including invalid timestamp.

Radar equations and measurement interpretation

When GEMSTAR Radar reflection TOF represents a round-trip path, R = 299,792,458 * TOF / 2. Conversion of reconstructed local axes uses (x_UE, -y_UE, z_UE) before adding mount translation in OSI metres for tested zero rotation. Invalid TOF or strength is rejected; no probabilistic sensor reliability calibration is established.

Why the static displacement is 2.008 m, not 2.000 m

The target translates 2 m along world X. Central rays are not perfectly axial; the distance along a slanted ray increases slightly more than 2 m. The 0.58 mm difference tests consistency between two simulated representations, not a real Radar manufacturer specification. The absolute baseline offset (~2.48 mm) and displaced offset (~3.06 mm) should be acknowledged.

Illustrated figure-by-figure interpretation

Every figure below is embedded from the supplied archive. Captions explain the research question, what the figure can support, and any key interpretation limit. Images are evidence from historical subtasks, not independently recreated measurements.

Tasks 23.5A-C - Radar returns, startup and reconstruction

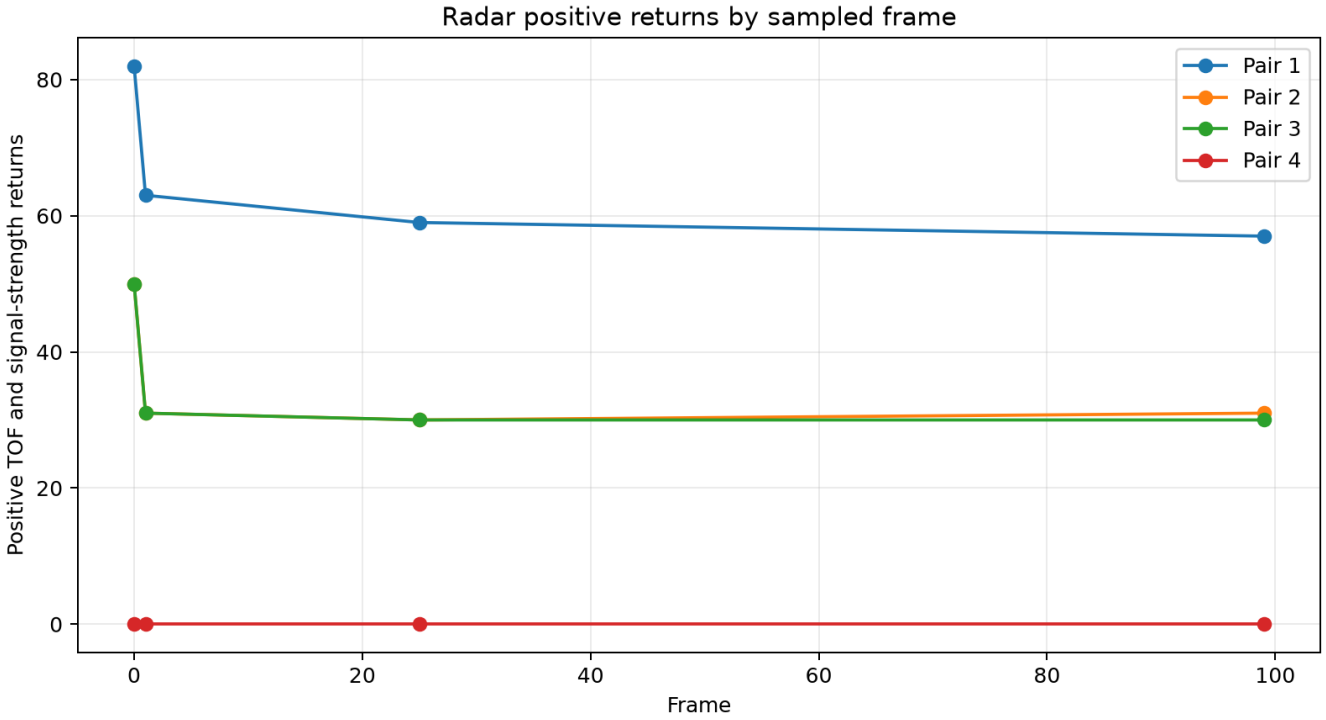


Figure 1. task23_05c1_radar_positive_returns.png - Counts positive Radar reflections in sampled frames; reflects usable returns, not object precision.

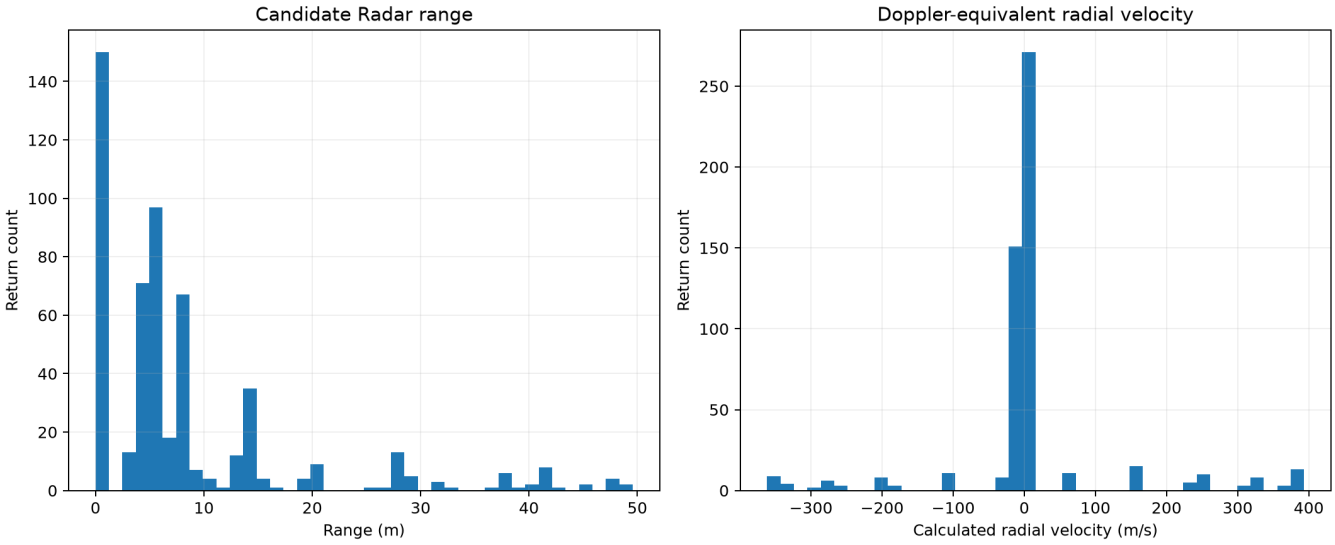


Figure 2. task23_05c1_radar_range_doppler.png - Plots Radar range and radial-velocity-related fields; startup artifacts and object identity require separate verification.

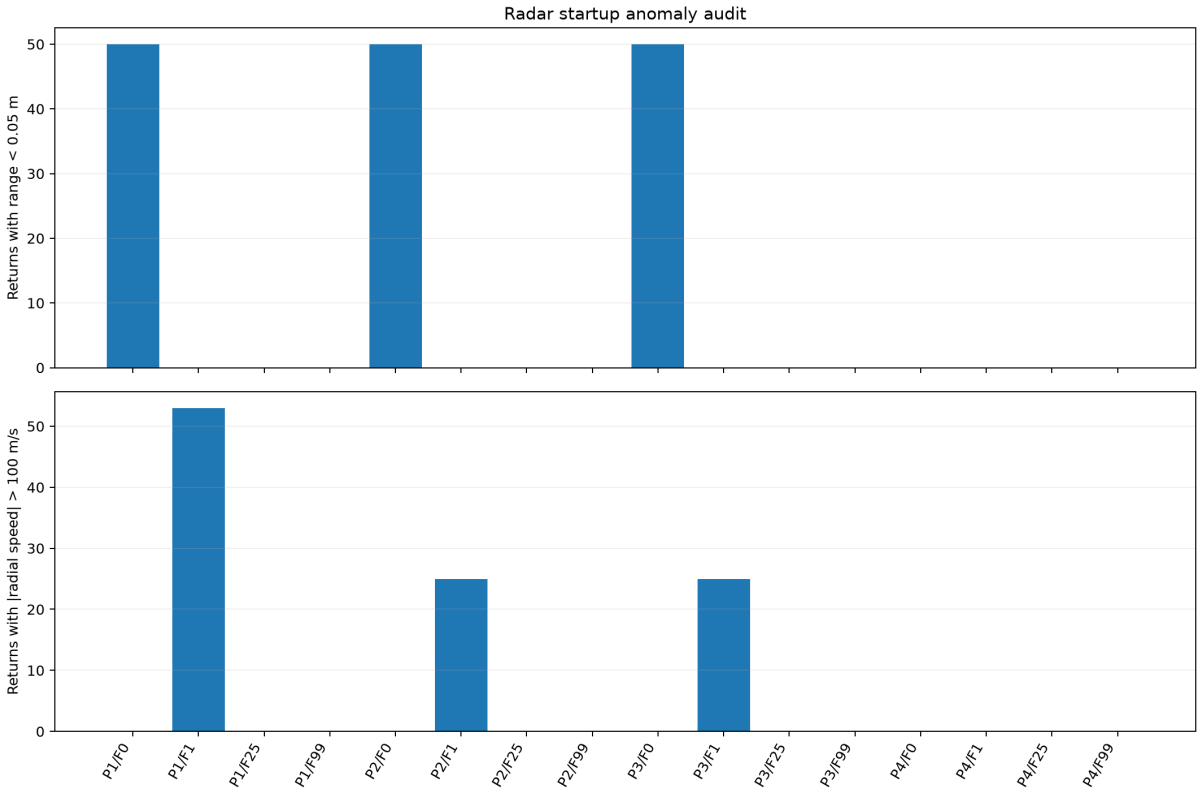


Figure 3. task23_05c2_radar_startup_anomalies.png - Shows early-frame Radar anomalies; avoid inferring steady-state behavior from startup frames.

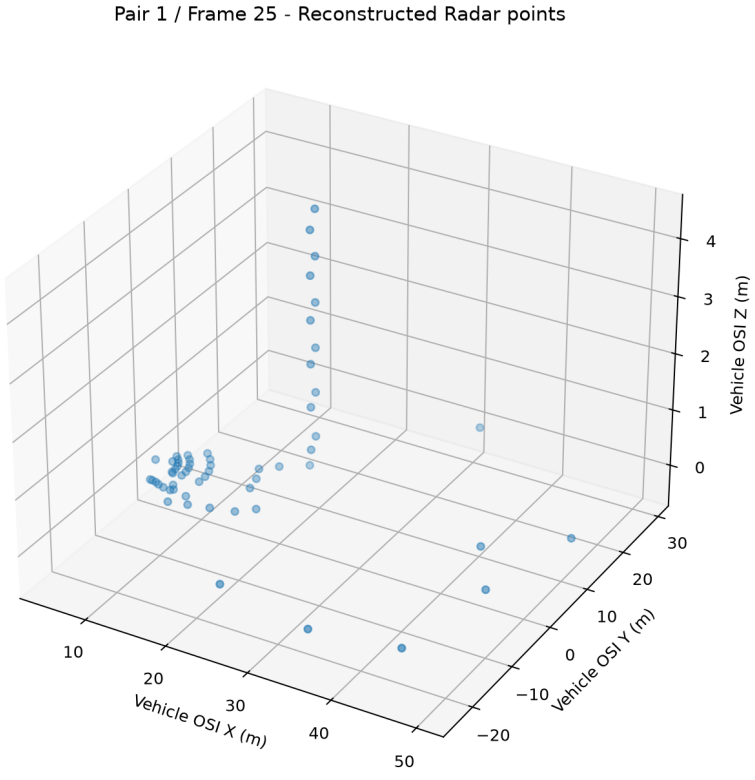


Figure 4. task23_05c3_radar_3d_pair1_frame25.png - Radar reflection cloud reconstructed at a sampled frame; geometry assumes the tested zero-rotation mount.

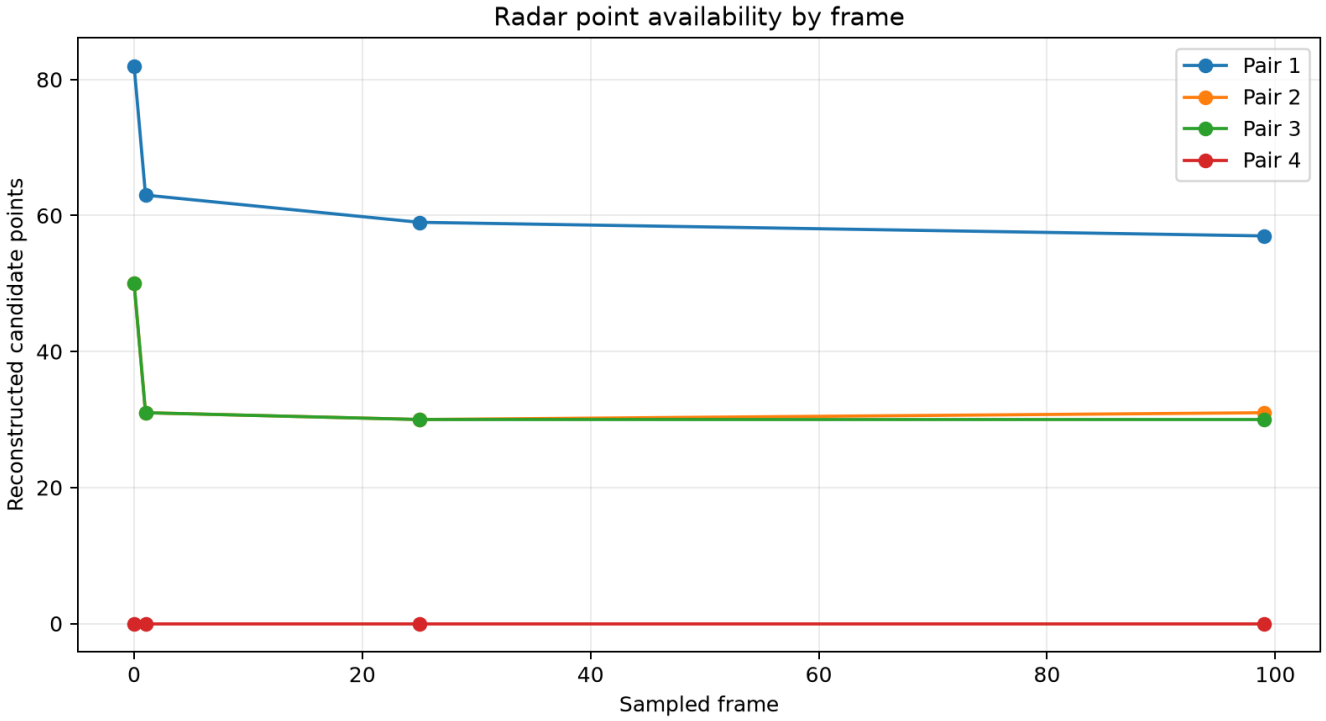


Figure 5. task23_05c3_radar_point_counts.png - Radar point counts across selected frames/recordings; assesses recording variability.

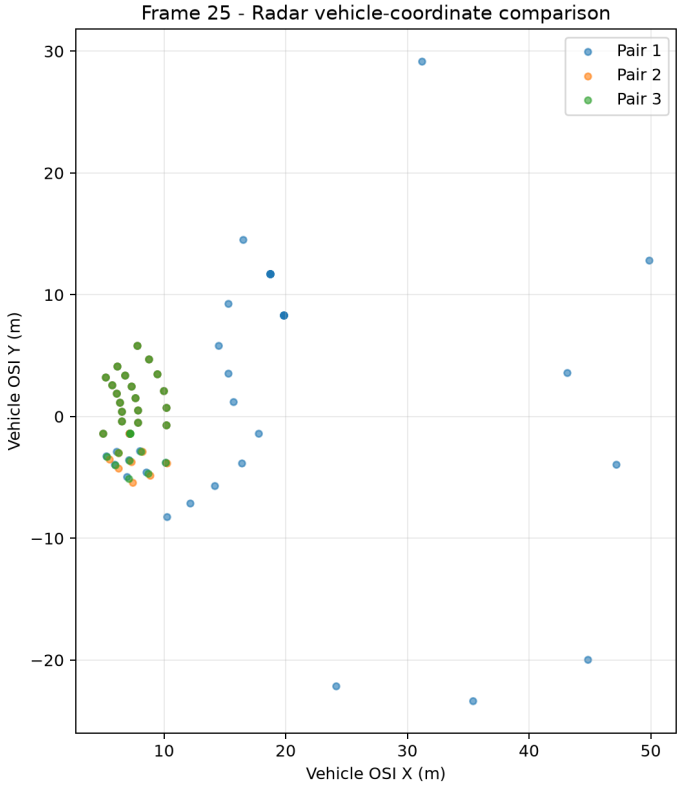


Figure 6. task23_05c3_radar_xy_pair_comparison.png - Overlay of Radar XY distributions from paired recordings; geometry alone does not verify object pairing.

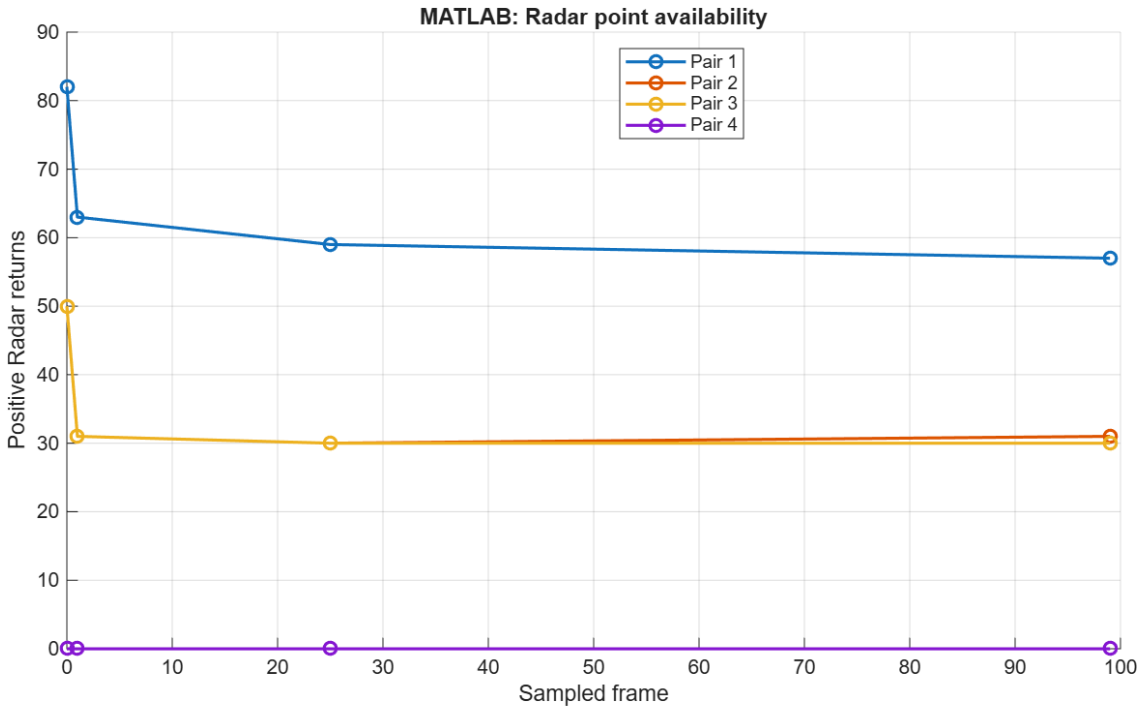


Figure 7. task23_05c4_matlab_radar_point_counts.png - MATLAB/Python Radar sample-count comparison for consistency across tools.

Tasks 23.5D - Spatial overlays and tentative candidate association

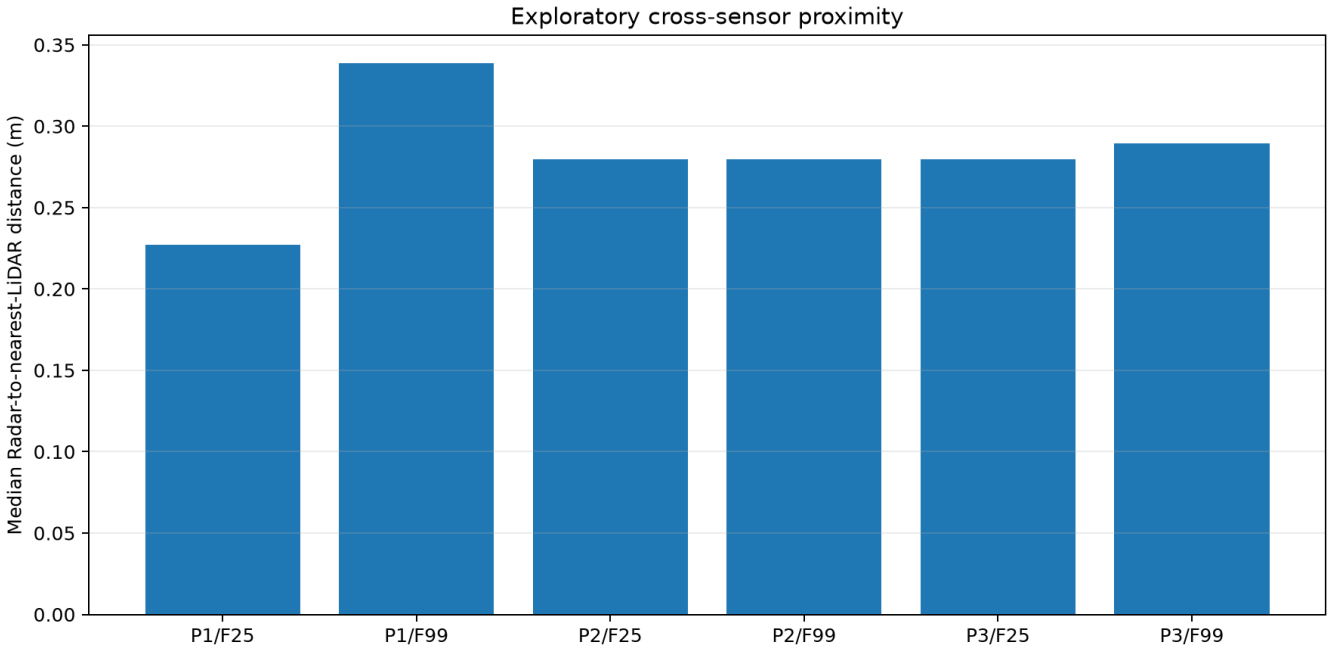


Figure 8. task23_05d1_nearest_distance_summary.png - Nearest Radar-to-LiDAR point distances; exploratory geometric overlap is not an object-association accuracy metric.

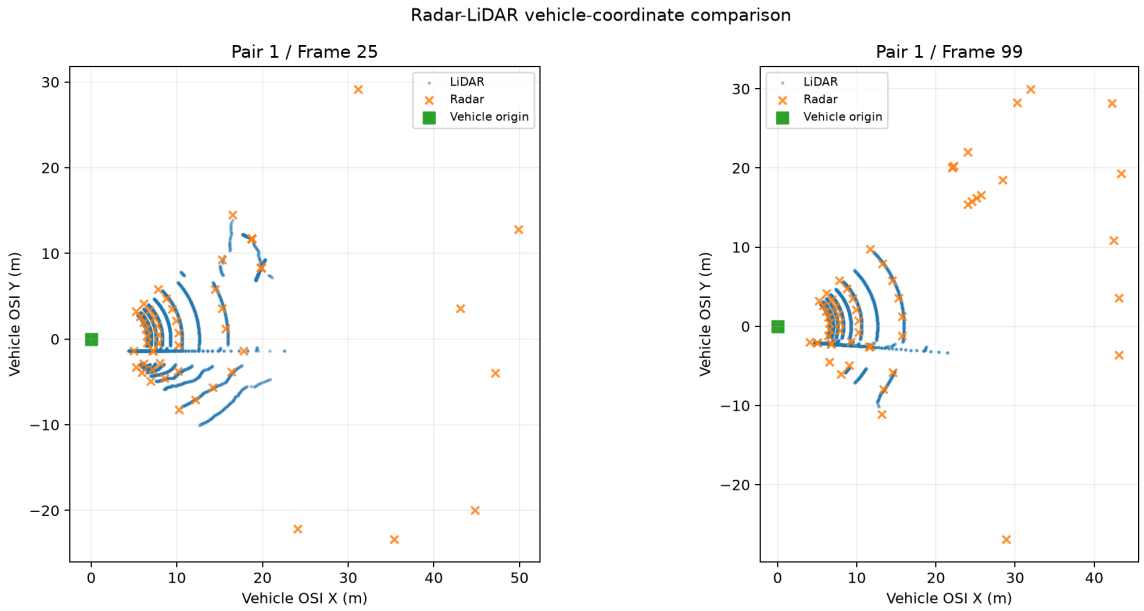


Figure 9. task23_05d1_pair1_radar_lidar_overlay.png - Radar and LiDAR points overlaid in the common configured vehicle frame; check outliers, occlusion and coverage mismatch.

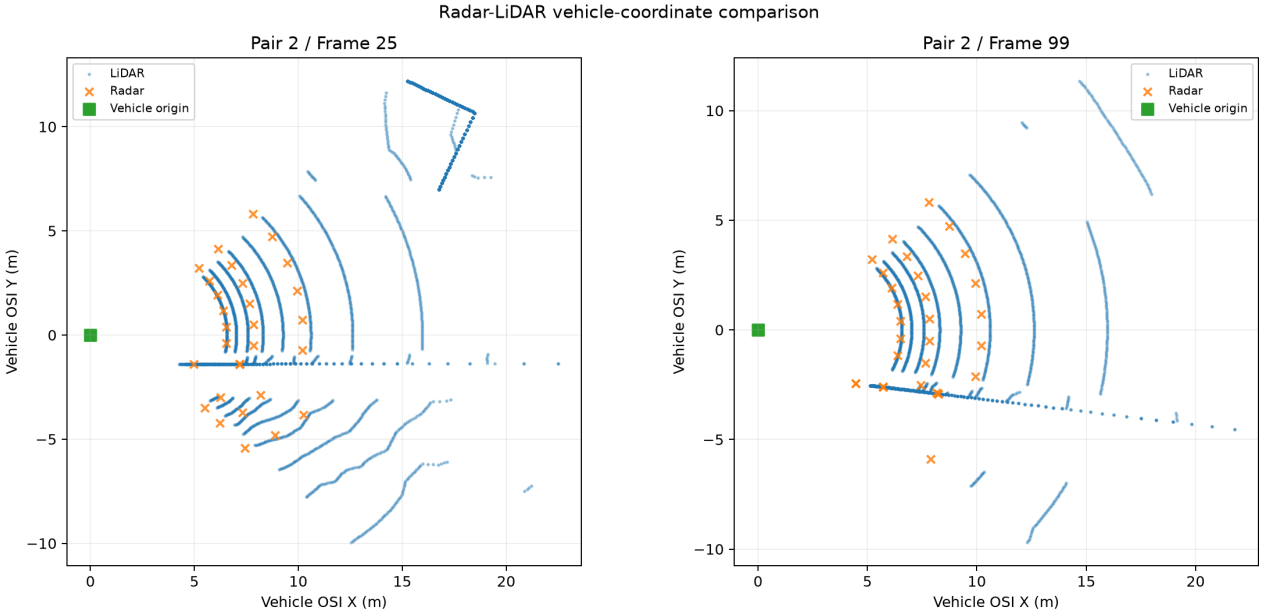


Figure 10. task23_05d1_pair2_radar_lidar_overlay.png - Radar and LiDAR points overlaid in the common configured vehicle frame; check outliers, occlusion and coverage mismatch.

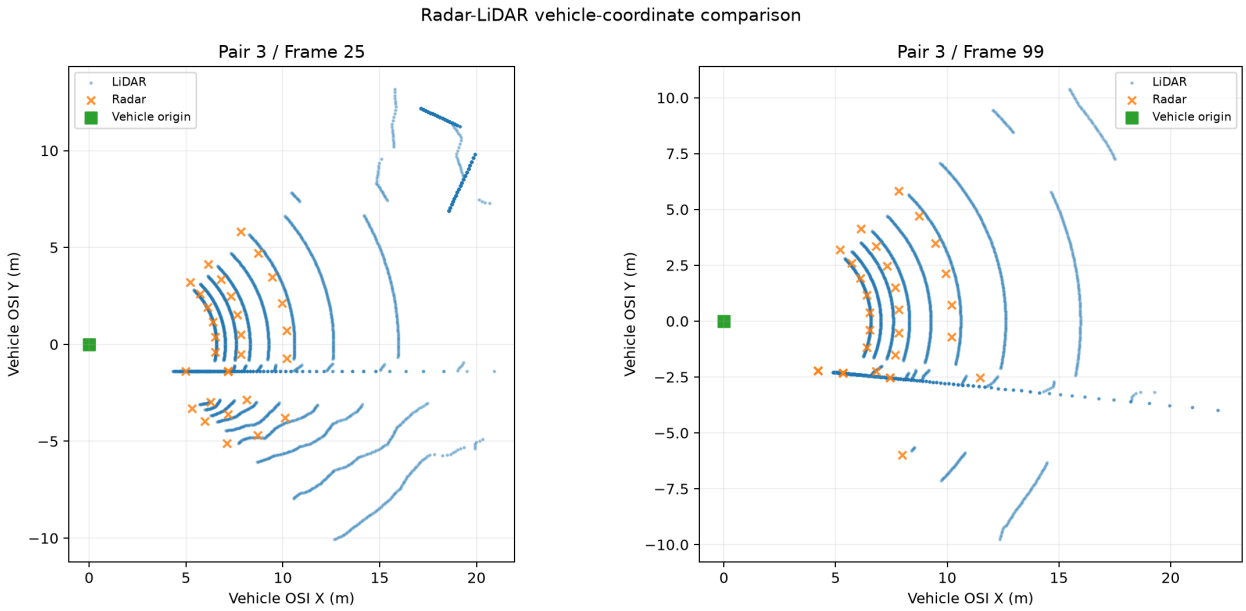


Figure 11. task23_05d1_pair3_radar_lidar_overlay.png - Radar and LiDAR points overlaid in the common configured vehicle frame; check outliers, occlusion and coverage mismatch.

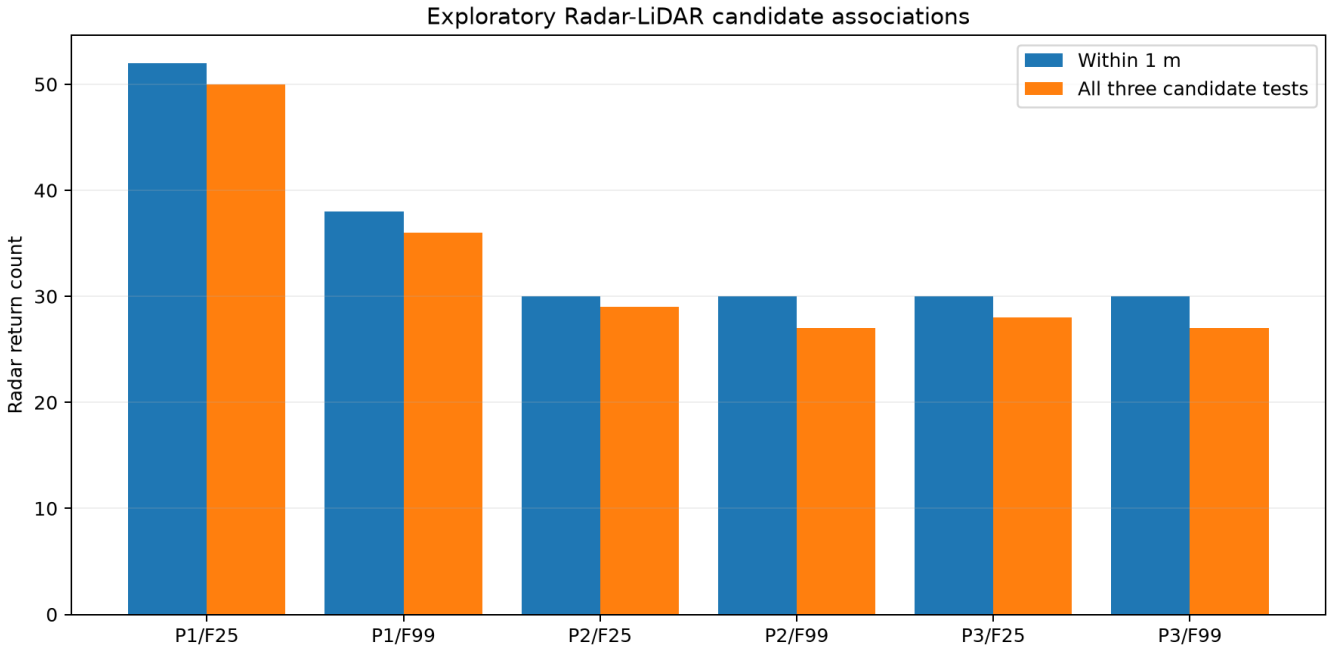


Figure 12. task23_05d3_candidate_association_counts.png - Number of candidate associations meeting a provisional geometric rule; candidates are not validated matches.

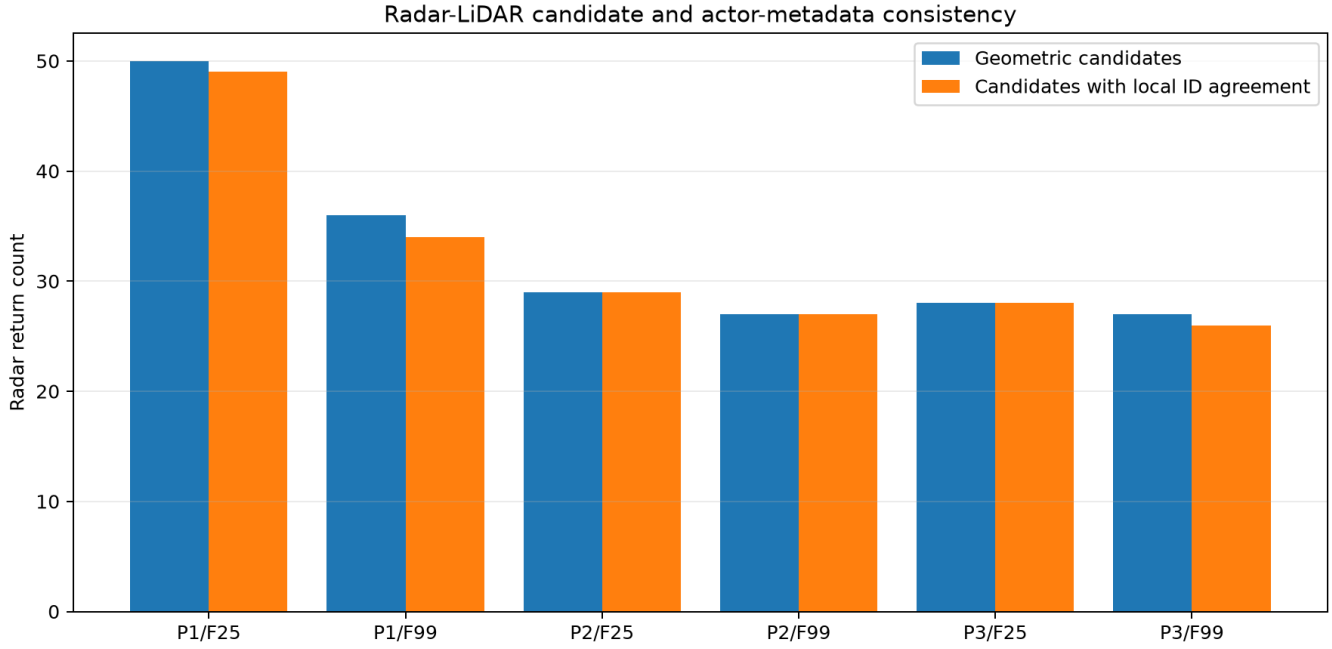


Figure 13. task23_05d4_actor_association_consistency.png - Checks agreement of selected candidate associations with actor references; requires independent identity evidence to claim correctness.

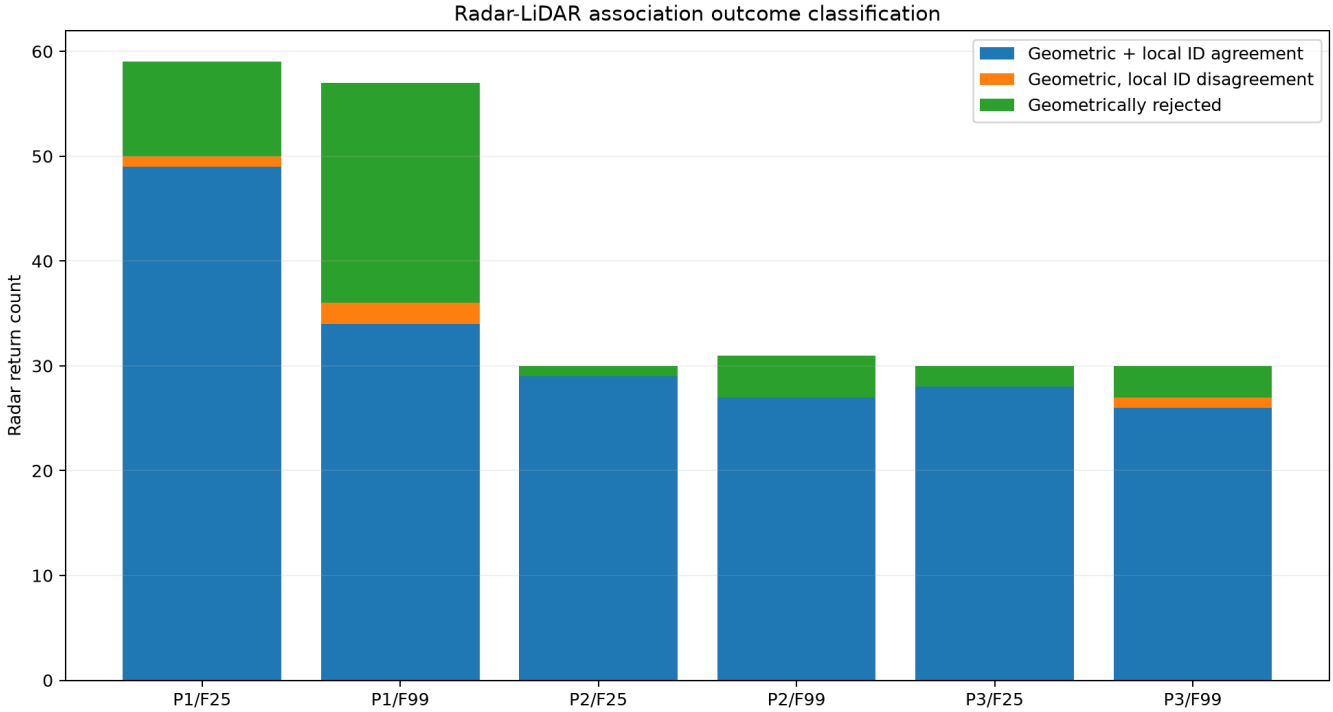


Figure 14. task23_05d5_association_outcomes.png - Classification of tentative association outcomes; interpret with archived rejection criteria.

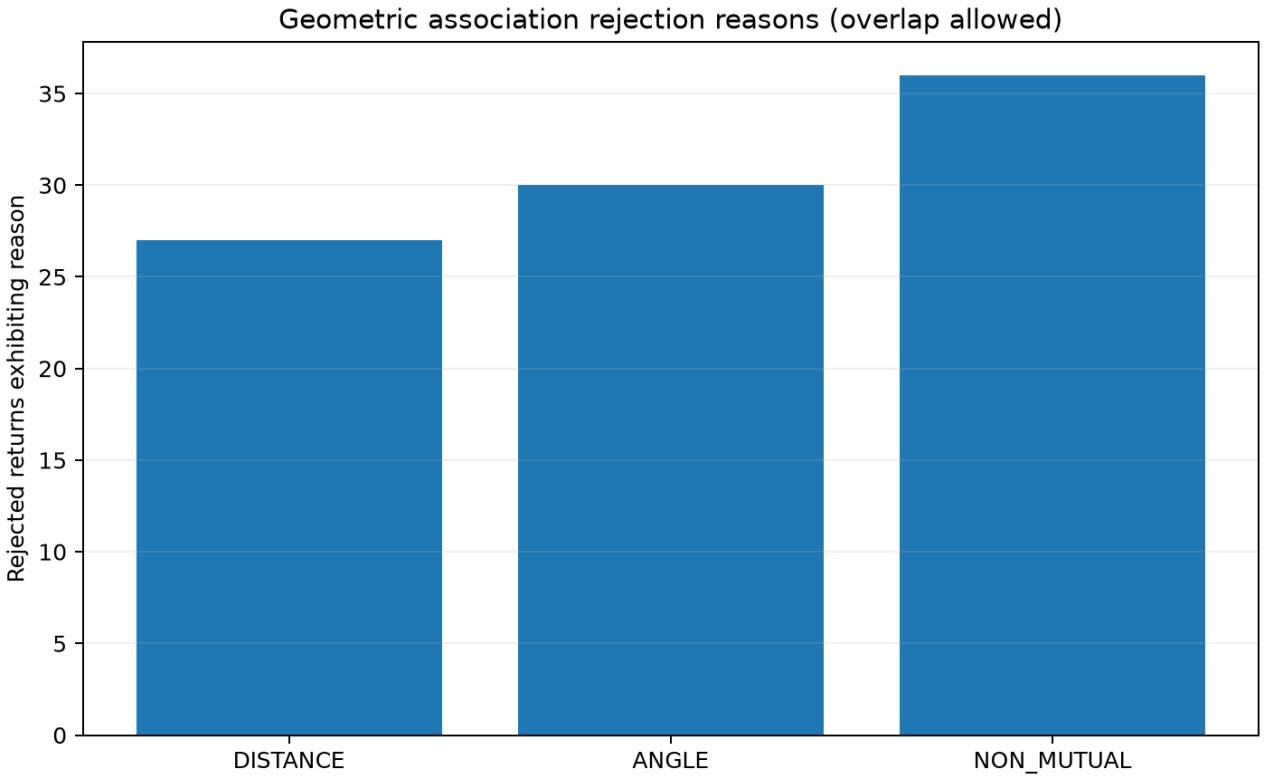


Figure 15. task23_05d5_rejection_reasons.png - Reasons why provisional association candidates were rejected; diagnostic for threshold and quality rules.

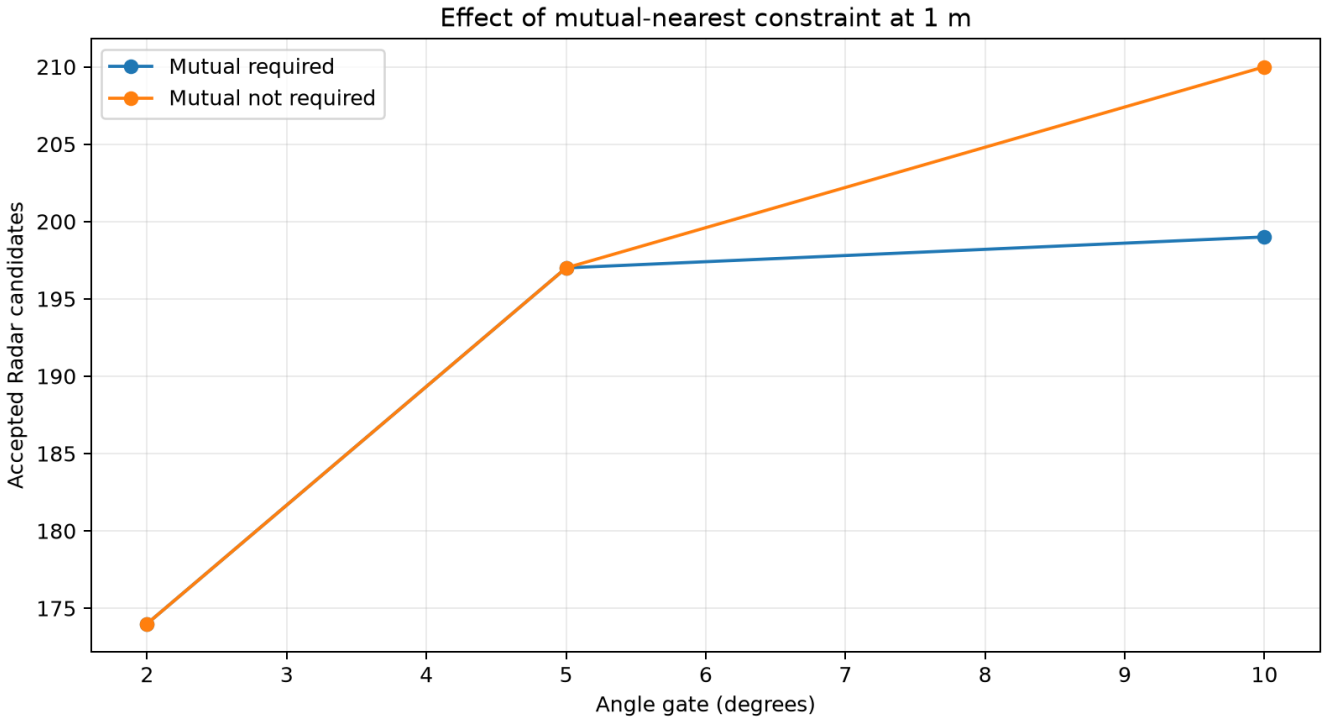


Figure 16. task23_05d6_mutual_constraint_effect.png - Effect of stricter mutual association checks; exploratory sensitivity, not detection correctness.

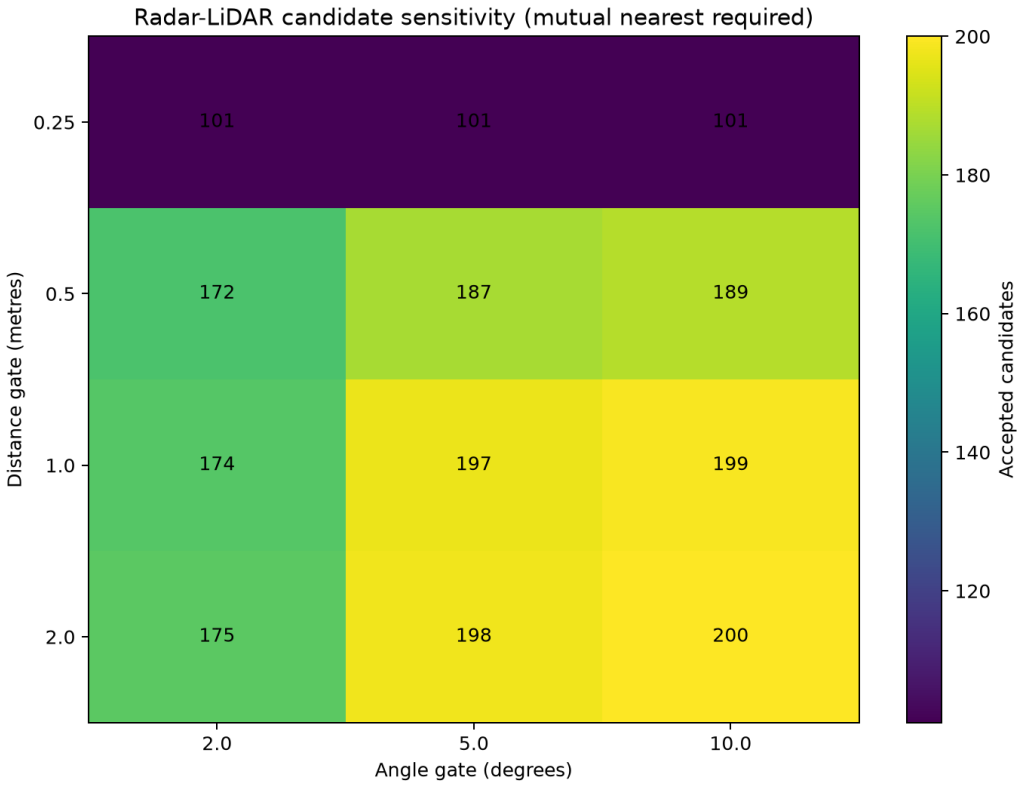


Figure 17. task23_05d6_threshold_sensitivity.png - Candidate association response to threshold selection; parameter sensitivity, not calibrated matching accuracy.

Task 23.13 - serialized Radar range comparison

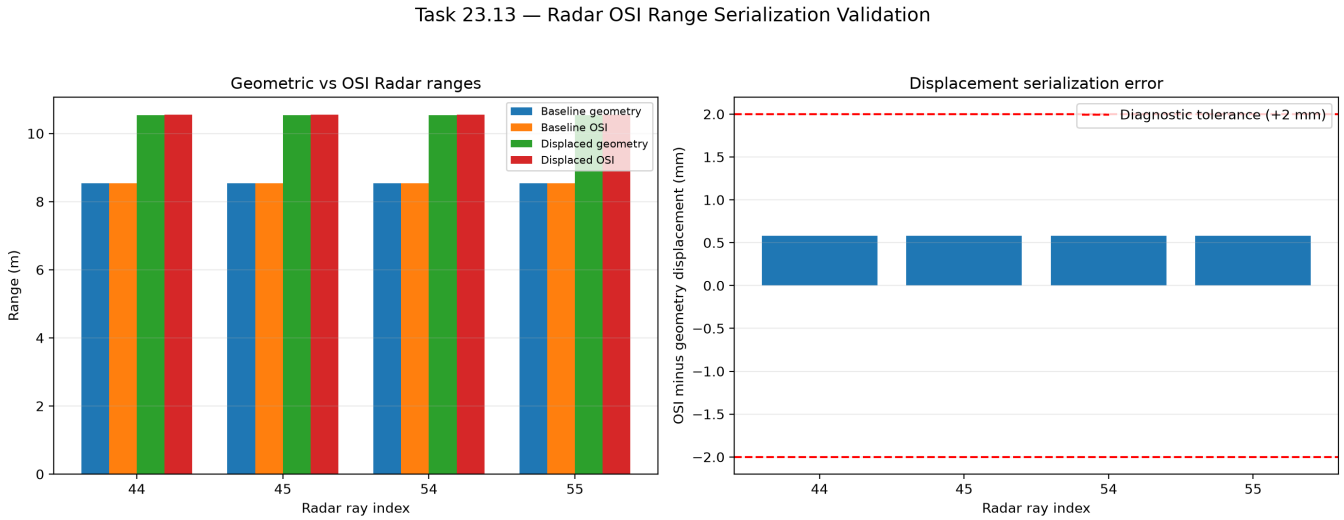


Figure 18. task23_13C_02_radar_serialization_comparison.png - Compares independent geometric Radar ranges with OSI-decoded TOF range and displacement for four static rays; tight agreement does not imply physical Radar accuracy.

Detailed subtask and code inventory

The table below is organized by archived script name and research objective. A filename identifies work that exists; it is not by itself a claim that every planned experiment or verification was completed.

Task 0 - source scripts (12)

task23_07af_measure_radar_ranges.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_07ag_geometry_validation.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_07ah_move_target.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_07ai_displacement_validation.py - Controlled change to target geometry for calibration.

task23_07aj_full_radar_comparison.py - Compare two measurements/scenarios using specified method.

task23_07ak_map_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_07ap_trace_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_07ar_move_and_verify.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_07ar_move_persistent.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_07ar_verify_persistent.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_07ar_verify_saved_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_07as_finalize_radar.py - Consolidate evidence and create a research-ready result.

Task 05a - source scripts (1)

task23_05a_radar_configuration_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

Task 05c - source scripts (4)

task23_05c1_radar_measurements.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_05c2_radar_startup_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_05c3_radar_3d_reconstruction.py - Reconstruct geometry, range, or Cartesian coordinates.

task23_05c4_matlab_radar_verification.m - MATLAB cross-tool calculation/plot based on exported research data.

Task 05d - source scripts (6)

task23_05d1_radar_lidar_alignment.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_05d2_radar_lidar_timestamp_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_05d3_candidate_associations.py - Explore correspondence candidates and rejection or identity assumptions.

task23_05d4_actor_association_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_05d5_association_failure_analysis.py - Explore correspondence candidates and rejection or identity assumptions.

task23_05d6_association_robustness.py - Explore correspondence candidates and rejection or identity assumptions.

Task 06d - source scripts (2)

task23_06d8i_remove_radar_tester.py - Run defined behavior or invalid-input checks.

task23_06d8j_verify_radar_actors.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 07 - source scripts (1)

task23_07_automated_calibration.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 07g - source scripts (1)

task23_07g_inspect_sensor_managers.py - Inspect recorded or simulated payloads before accepting analysis.

Task 07r - source scripts (1)

task23_07r_validate_osi.py - Test or numerical validation of a prior dataset/implementation.

Task 07s - source scripts (2)

task23_07s_decode_radar.py - Interpret encoded Camera/Radar/OSI measurements.

task23_07s_decode_radar_before_fix.py - Interpret encoded Camera/Radar/OSI measurements.

Task 07u - source scripts (1)

task23_07u_reflection_inspection.py - Inspect recorded or simulated payloads before accepting analysis.

Task 07v - source scripts (1)

task23_07v_radar_configuration.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 07w - source scripts (1)

task23_07w_reference_geometry.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 07y - source scripts (1)

task23_07y_align_radar_target.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 13B - source scripts (2)

task23_13B_02_decode_radar_calibration.py - Interpret encoded Camera/Radar/OSI measurements.

task23_13B_03_validate_radar_serialization.py - Test or numerical validation of a prior dataset/implementation.

Task 13C - source scripts (2)

task23_13C_01_radar_robustness_tests.py - Run defined behavior or invalid-input checks.

task23_13C_02_radar_comparison_plot.py - Generate visual research evidence from previously recorded data.

Task 13D - source scripts (1)

task23_13D_01_finalize_radar.py - Consolidate evidence and create a research-ready result.

Task 15 - source scripts (1)

task23_15_radar_adapter.py - Wrap validated sensor-specific operation into common interface.

CSV evidence and reproducibility register

CSV file	Rows	Example fields
task23_05a_radar_raw_measurement_summary.csv	16	PairID, Frame, SensorIndex...
task23_05c1_radar_frame_summary.csv	16	PairID, Frame, TotalSlots...
task23_05c1_radar_measurements.csv	1600	PairID, Frame, RayIndex...

CSV file	Rows	Example fields
task23_05c2_radar_startup_quality_audit.csv	16	PairID, Frame, Category...
task23_05c3_radar_3d_points.csv	544	PairID, Frame, QualityCategory...
task23_05c3_radar_3d_validation.csv	16	PairID, Frame, QualityCategory...
task23_05c4_matlab_python_comparison.csv	16	PairID, Frame, PositivePoints...
task23_05d1_radar_lidar_alignment.csv	6	PairID, Frame, RadarPoints...
task23_05d2_radar_lidar_timestamps.csv	15	PairID, Frame, TargetFrame...
task23_05d2_timestamp_progression.csv	9	PairID, FromFrame, ToFrame...
task23_05d3_candidate_association_details.csv	237	PairID, Frame, RadarRayIndex...
task23_05d3_candidate_association_summary.csv	6	PairID, Frame, RadarPoints...
task23_05d4_actor_association_details.csv	237	PairID, Frame, RadarRayIndex...
task23_05d4_actor_association_summary.csv	6	PairID, Frame, RadarPoints...
task23_05d4_actor_candidate_counts.csv	20	PairID, Frame, LiDARActorID...
task23_05d5_association_classification.csv	237	PairID, Frame, RadarRayIndex...
task23_05d5_candidate_actor_distribution.csv	20	PairID, Frame, AssignedNearestLiDARActorID...
task23_05d5_failure_cases.csv	44	PairID, Frame, RadarRayIndex...
task23_05d5_frame_summary.csv	6	PairID, Frame, RadarReturns...
task23_05d6_frame_actor_concentration.csv	6	PairID, Frame, RadarReturns...

CSV file	Rows	Example fields
task23_05d6_threshold_sensitivity.csv	24	DistanceGateM, AngleGateDeg, RequireMutual...
task23_06b_actor_source_matches.csv	131	SourcePath, LineNumber, SourceLine
task23_06b_relevant_unreal_assets.csv	1	Name, FullName, Length
task23_06b_scenario_inventory.csv	38	Name, Extension, SizeBytes...
task23_07_reference_target.csv	1	TargetLabel, RadarXcm, RadarYcm...
task23_07af_radar_range_measurements.csv	9572	Frame, RadarView, RayIndex...
task23_07ag_geometric_comparison.csv	9572	Frame, RayIndex, MeasuredRange_m...
task23_07ai_displacement_comparison.csv	4	Ray, BaselineSamples, MovedSamples...
task23_07aj_all_rays_comparison.csv	95	RayIndex, BaselineRange_m, MovedRange_m...
task23_07as_radar_displacement_final.csv	4	ray, baseline_distance_cm, displaced_distance_cm...

CSV file	Rows	Example fields
task23_07j_recording_inventory.csv	0	
task23_07r_osi_structure_validation.csv	2	File, Status, Messages...
task23_07s_radar_frames.csv	1534	Frame, TimestampSeconds, TimestampNanos...
task23_13B_02_radar_calibration_ranges.csv	80	Scenario, FileName, Frame...
task23_13B_03_radar_serialization_comparison.csv	4	Ray, BaselineGeometryM, BaselineOSIM...
task23_13C_01_radar_robustness_tests.csv	12	Test, ExpectedAccept, ActualAccept...

Missing figures, why they matter, and how to create them

Figure / experiment	Why it is needed	Source / feasibility
Four-ray range residual plot in millimetres	Makes small systematic geometric-vs-OSI differences obvious	Can generate from task23_13B_03_radar_serialization_comparison.csv.
Radar range/Doppler on moving target	Static cube cannot validate radial speed, tracking or startup dynamics	Requires controlled known-speed driving/target experiment.
Clutter/occlusion/fog robustness plot	Necessary for scene-dependent Radar usefulness	Not established by static calibrated rays; require appropriate matched scenarios.
Validated actor association plot	Candidate association counts are not precision/recall	Requires independent GT identities and synchronized 3-sensor data in Task 24.

Scientific interpretation and handover

The checked results are a reproducible software and simulator preprocessing baseline. They do not establish real-world accuracy, dynamic multi-object perception, CPU-versus-FPGA energy improvement, or measured deadlines. Those claims require separate matched scenarios, explicit ground-truth evaluation, and physical hardware benchmarking.

All embedded plots/images are original archived figures except where a figure is explicitly described as a newly synthesized summary. The archived source file name accompanies each caption. Raw multi-gigabyte OSI sources are not embedded and were not reprocessed for these PDFs.